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AI-Assisted Cartonization and Packing Optimization System for E-Commerce Logistics

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Abstract: - Efficient cartonization and packing play a critical role in reducing operational costs, material waste, and transportation inefficiencies in e-commerce logistics. Traditional rule-based packing systems often fail to adapt to diverse product dimensions, dynamic order compositions, and scalability requirements of modern fulfilment centres. This paper proposes an AI-assisted cartonization and packing optimization system designed to intelligently select optimal carton sizes and packing configurations for e-commerce orders. The proposed system integrates machine learning techniques for carton size prediction with optimization-based packing algorithms to maximize space utilization while ensuring item safety constraints. A modular system architecture is developed using a service-oriented backend, enabling seamless integration with order management systems and visualization interfaces. Experimental evaluation conducted on representative e-commerce order datasets demonstrates a significant reduction in unused carton volume, packaging material consumption, and overall shipping costs when compared to conventional heuristic-based approaches. The results highlight the effectiveness of combining artificial intelligence with optimization strategies for scalable and adaptive cartonization in real-world logistics environments. The proposed framework offers a practical and extensible solution for intelligent warehouse fulfilment operations.

Keywords: AI-assisted cartonization, packing optimization, e-commerce logistics, 3D bin packing, machine learning, warehouse automation

1. INTRODUCTION

The rapid growth of e-commerce has markedly increased the complex nature of logistics and warehouse fulfilment processes. Among these operations, cartonization and packing play a crucial role in determining shipping efficiency, material consumption, and transportation costs. Inefficient carton selection often leads to excessive void space, increased use of packaging materials, higher shipping charges, and a larger environmental footprint. As order profiles become increasingly diverse in terms of product dimensions, weights, and combinations, traditional packing strategies struggle to deliver optimal performance. Conventional cartonization systems predominantly rely on rule-based heuristics and fixed carton size catalogs. While these approaches are computationally efficient, they lack adaptability to dynamic order characteristics and do not scale well with large and heterogeneous product inventories. As a result, such systems

frequently produce suboptimal packing configurations, leading to underutilized carton space and increased logistics costs. These limitations highlight the need for intelligent cartonization solutions capable of learning from historical data and adapting to evolving operational conditions. Recent advancements in artificial intelligence and optimization techniques have opened new possibilities for addressing complex combinatorial problems in logistics. Machine learning models can analyze historical order and packing data to predict suitable carton sizes, while optimization algorithms can determine efficient item placement strategies within selected cartons. Integrating these techniques enables the development of intelligent packing systems that outperform traditional heuristic-based methods in terms of space utilization and cost efficiency. This paper presents an **AI-assisted cartonization and packing optimization system for e-commerce logistics** that combines machine learning-based carton size prediction with optimization-driven packing strategies. The proposed system is designed using a modular and scalable architecture, allowing seamless integration with existing order management and warehouse systems. By intelligently selecting carton sizes and optimizing item placement, the system aims to minimize unused carton volume, reduce packaging material usage, and lower overall shipping costs. Experimental evaluation demonstrates that the proposed system achieves improved packing performance when compared to conventional heuristic-based cartonization methods. Conventional cartonization systems predominantly rely on rule-based heuristics and fixed carton size catalogs. While these approaches are computationally efficient, they lack adaptability to dynamic order characteristics and do not scale well with large and heterogeneous product inventories. Further sections of this paper are organized as follows. The proposed system's user-friendliness and warehouse utility are discussed in Section II. Section III reviews current cartonization and packing optimization literature. The proposed system architecture and methodology are in Section IV. Section V details system implementation. System architecture and methodology are described in Section VI. Section VII details system implementation. Section VI reports experimental results and performance analysis, and Section VII concludes with a summary and future research.

II.EASE OF USE

I. Artificial Intelligence–Based Carton Size Prediction

Artificial intelligence methodologies significantly facilitate the usability of the proposed cartonization system through the automation of carton size selection. In the present work, machine learning models are utilized to analyze historical order data, including item dimensions, weight, and quantity, in order to predict the most suitable carton size for each order. This automated prediction eliminates the need for manual carton selection and reduces dependency on predefined rule-based strategies. Due to its data-driven nature, the system adapts to varying order profiles and enables faster and more accurate cartonization decisions, thereby improving operational efficiency in e-commerce fulfillment centers.

II. Optimization-Based Packing Strategy

To further improve usability and packing efficiency, optimization algorithms are employed to determine effective item placement within the selected cartons. These algorithms ensure maximum space utilization while maintaining item safety constraints. By automatically generating optimized packing configurations, the system minimizes unused carton volume and reduces packaging material waste. The integration of optimization techniques enhances system reliability and consistency, allowing users to obtain optimal packing results without requiring expertise in packing algorithms or logistics planning.

III. User-Friendly Application Interface

The proposed system is designed with a user-friendly interface that supports seamless interaction for warehouse operators and logistics personnel. The application enables users to input or retrieve order details and receive immediate carton recommendations along with optimized packing layouts. Clear visualization and simplified output formats ensure that users can easily interpret the system's recommendations without extensive technical knowledge. By reducing manual effort and providing real-time decision support, the system improves usability and facilitates practical deployment in real-world e-commerce logistics environments.

III. LITERATURE SURVEY

Duan et al. [1] introduced a multi-task selected learning framework to address the three-dimensional flexible bin packing problem (BPP) in extensive e-commerce warehouses. The approach uses reinforcement learning to simultaneously generate packing sequences and item orientations with the objective of minimizing bin surface area. By balancing task correlation through selected learning, the model learns heuristic-like packing policies without explicit heuristic design. Experimental results on real-world transaction datasets show that the proposed method outperforms traditional heuristic and learning-based approaches, achieving an average surface area cost reduction of 5.47% compared to production-level greedy

The study in [2] investigates a three-dimensional bin packing problem involving irregular-shaped and deformable items, with a focus on applications in city logistics and fresh food delivery. The authors present a constructive heuristic that utilizes an improved dynamic-volume bottom-left-fill strategy, integrating item compression and stability constraints. The algorithm enhances load stability and space utilization by prioritizing items with larger bottom facets and greater compressibility. Experimental results show that considering item deformation enables higher packing efficiency compared to conventional rigid-item packing approaches.

Ananno and Ribeiro [3] developed a two-stage multi-heuristic algorithm for multi-container three-dimensional bin packing under industrial constraints. The methodology integrates constructive heuristics to create compact product layers with a genetic algorithm for the efficient packing of residual items. The method is assessed using real-world industrial and academic benchmark datasets, exhibiting consistent performance across both homogeneous and heterogeneous order profiles. Results indicate improved container utilization, compactness, and load stability, contributing to reduced packaging material usage and enhanced sustainability.

The research in [4] introduces a deep reinforcement learning framework designed to address the online three-dimensional bin packing problem in dynamic settings characterized by the sequential arrival of items. The proposed approach integrates reinforcement learning with stability constraints and object rearrangement to improve packing flexibility and space utilization. Experimental results demonstrate that the model outperforms traditional heuristic and learning-based methods in terms of packing density and structural stability. The results demonstrate the suitability of deep reinforcement learning for real-time logistics system dynamic packing optimization.

Xiong et al. [5] compared deep reinforcement learning and traditional heuristic methods for online three-dimensional bin packing in robotic packing. The study treats packing as a Markov decision process and proposes a DRL framework with a candidate map to reduce action space and ensure placement feasibility and stability. Experimental results show that the proposed DRL-based approach outperforms traditional heuristics, multi-objective optimization strategies, and baseline DRL models in terms of space utilization. The authors also discuss practical limitations of DRL methods and provide insights for real-world deployment in logistics and manufacturing systems.

Xiong et al. [6] proposed a transformer-based deep reinforcement learning framework, termed GOPT, for solving the online three-dimensional bin packing problem with enhanced generalization capability. The approach introduces a placement generator to produce feasible packing sub-spaces and a packing transformer that uses cross-attention to model spatial correlations between items and available bin regions. Experimental results demonstrate that GOPT outperforms existing DRL-based methods while generalizing effectively across varying bin dimensions and unseen items. The

effective implementation of the trained policy on a robotic system further confirms the practical relevance of the proposed method in real-world logistics and automation contexts.

Zhao and Yang [7] studied order batching in multi-cross-aisle e-commerce warehouses to improve order-picking efficiency. The study proposes a tabu search-based batching algorithm combined with a dynamic programming-assisted picker routing strategy to optimize batch formation and travel distance. Experimental evaluation using real-world warehouse data demonstrates that the proposed approach significantly improves order-picking performance compared to existing operational practices. The findings emphasize that effective order batching strategies have a greater impact on warehouse efficiency than routing rules alone.

Jia et al. [8] developed a robotic online three-dimensional bin packing strategy for intelligent palletizing using deep reinforcement learning and 3D vision. Point-cloud-based object size and pose estimation are combined with deep reinforcement learning and Monte Carlo tree search to manage large action spaces. Experimental findings indicate that the acquired policy achieves higher space utilization and robustness relative to conventional heuristic approaches, while adhering to real-time operational constraints. The research confirms the practical viability of DRL-based online packing strategies for actual robotic logistics systems.

Romero et al. [9] developed Q4RealBPP, a hybrid quantum-classical framework for logistics-oriented bin packing under realistic industrial constraints. The approach extends classical bin packing formulations by incorporating heterogeneous bins, delivery priorities, and item-bin association requirements, while supporting one-, two-, and three-dimensional problem instances. Experimental evaluation demonstrates the capability of the quantum-assisted framework to handle complex, real-world packing scenarios effectively. The study highlights the potential of hybrid quantum optimization techniques as an emerging alternative for addressing large-scale logistics optimization problems.

Mao et al. [10] employed deep learning to continuously choose the most effective heuristic for variable-sized bin packing. The study models bin characteristics and packing performance using large-scale real-world data and automatic feature learning. Experimental results on millions of instances demonstrate that the proposed model achieves high prediction accuracy and effectively improves bin packing solution quality. The work highlights the effectiveness of data-driven heuristic selection for large-scale industrial bin packing problems.

Krisadee and Jindaluang [11] presented an improved heuristic solution for the three-dimensional BPP with the objective of minimizing the number of bins while satisfying spatial and weight constraints. The study enhances an existing three-stage layer-based heuristic by retaining only placement patterns that maximize volume utilization,

thereby reducing computational overhead. Experimental results show significant reductions in bin usage, execution time, and memory consumption, along with improved space utilization efficiency. The work demonstrates that refining classical heuristic strategies can yield substantial performance gains for large-scale 3D packing problems.

Kundu et al. [12] introduced a deep reinforcement learning approach, named Deep-Pack, to address the online two-dimensional BPP utilizing a vision-based state representation. The approach utilizes a Double DQN framework that interprets the current bin state as an image and predicts optimal placement positions to enhance packing density. Experimental results show that the proposed method outperforms online 2D packing techniques with packing efficiencies above 90%. The study also shows that the vision-based DRL framework can be applied to robotic three-dimensional online bin packing scenarios.

Krisadee and Jindaluang [13] proposed an enhanced heuristic for the “three-dimensional bin packing problem (3D-BPP)” aimed at minimizing the number of bins while satisfying both spatial and weight constraints. Their method refines the classical Three-Stage Layer-Based Heuristic (TSLBH) by reducing redundant placement patterns and retaining only configurations that maximize volume utilization without violating bin capacity limits. This selective pattern storage significantly decreases computational overhead in later packing stages. Experimental evaluation on large-scale datasets (up to one million boxes) showed notable performance gains, including reductions of 38.68% in bin usage, 58.42% in execution time, and 63.33% in memory consumption, while improving space utilization efficiency by 35.38%. The study demonstrates that careful optimization of classical heuristic frameworks can produce substantial improvements in scalability and efficiency for real-world large-scale 3D packing applications.

Gong [14] proposed an artificial intelligence-driven optimization approach for logistics warehousing and distribution path planning, focusing on improving routing efficiency and reducing operational costs in intelligent logistics systems. The study introduces an enhanced swarm intelligence-based optimization algorithm to overcome the limitations of traditional routing methods that often converge prematurely to local optima. By integrating AI techniques such as intelligent data analysis, real-time route planning, and automated decision-making, the model supports dynamic vehicle routing under practical constraints like delivery time windows and transit frequency. Experimental comparisons demonstrate that the improved algorithm achieves faster convergence, better global search capability, and lower logistics distribution costs compared to conventional optimization methods. The work highlights the growing role of AI in enabling smart warehousing, automated transportation, and intelligent supply chain management, showing strong potential for large-scale real-world logistics applications.

N. Karthikeyan, R. Priyadharshini, and S. Mohanraj [15] introduced a predictive analytics framework for improving e-

20 commerce supply chains utilizing Categorical Boosting (CatBoost), a gradient boosting method adept at managing categorical and heterogeneous logistics data. The study focuses on improving demand forecasting, inventory planning, and distribution efficiency by leveraging machine learning to model complex nonlinear relationships in large-scale supply chain datasets. Unlike traditional statistical forecasting approaches, the proposed model automatically learns feature interactions and reduces preprocessing effort while maintaining high predictive accuracy. Experimental evaluation demonstrates improved forecasting precision, reduced stock imbalances, and better logistics decision-making performance compared to baseline machine learning models. The work highlights the growing importance of data-driven predictive modeling in enhancing responsiveness, cost efficiency, and scalability in modern e-commerce supply chain operations.

IV. PROPOSED SYSTEM

An AI-powered cartonization and packing optimization system is proposed to improve the efficiency of order fulfillment in warehouse and e-commerce logistics environments. Traditional packing systems rely heavily on static rules or manual decision-making, which often leads to poor space utilization, increased carton usage, and higher logistics costs. The proposed system integrates rule-based packing policies, inventory-aware allocation, and AI-assisted packing recommendations to achieve optimized carton selection and item grouping.

The proposed framework processes customer orders through a centralized packing engine that evaluates item dimensions, weight, quantity, and real-time inventory availability. Based on predefined packing policies such as master pack, inner pack, broken carton handling, and drop-ship constraints, the system determines the most suitable packing strategy. Additionally, an AI module leverages historical packing data to recommend efficient carton usage and improve space utilization.

The process begins when a user selects an order through the web-based dashboard. The order details are then passed to the packing engine, where items are validated against available inventory. The system applies packing rules and constraints to group items logically and selects suitable cartons that satisfy dimensional and weight limits. Once packing is completed, the system generates pick tickets, packing lists, UCC labels, and provides a 3D visualization of the packed cartons for verification.



Fig. 1. Proposed Framework of AI-Based Cartonization System

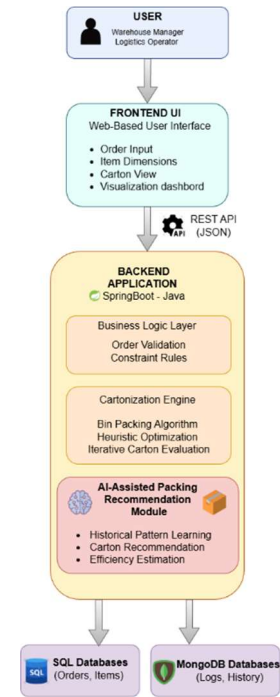


Fig. 2. System Architecture Diagram

A. Theoretical Background and Motivation

Cartonization and packing optimization issues are classified as NP-hard combinatorial optimization problems, closely associated with the traditional BPP and the 3D-BPP. Exact optimization techniques become computationally impractical for extensive, real-time warehouse operations because of the exponential increase in search space. Consequently, practical systems rely on heuristics, rule-based constraints, and approximation methods to achieve near-optimal solutions within acceptable execution time.

However, purely heuristic approaches often fail to adapt to dynamic order profiles, diverse item characteristics, and evolving logistics constraints. To address this limitation, the proposed system adopts a hybrid theoretical approach, combining deterministic policy-based decision logic with data-driven AI-assisted inference, enabling both reliability and adaptability.

B. Formal Mapping to Bin Packing Theory

The proposed cartonization problem is modeled as a constrained multi-dimensional bin packing problem, where items must be assigned to bins (cartons) under multiple constraints.

Let:

- $I = \{i_1, i_2, \dots, i_n\}$ be the set of items
- $C = \{c_1, c_2, \dots, c_m\}$ be the set of cartons

Unlike classical bin packing, the proposed formulation introduces domain-specific constraints, including:

1. Inventory availability
2. Policy-based grouping (master pack, inner pack)
3. Drop-ship handling
4. Real-time transactional consistency

Thus, the problem extends traditional BPP into a policy-

constrained operational optimization problem, making it more aligned with real-world warehouse systems.

C. Mathematical Formulation of the Packing Problem

13 Let an order consist of a finite set of items

$$I = \{i_1, i_2, \dots, i_n\}$$

Each item i_k is represented as:

3
$$i_k = (l_k, w_k, h_k, wt_k, q_k)$$

where l_k, w_k, h_k denote the item dimensions, wt_k denotes item weight, and q_k denotes the ordered quantity.

Let the available carton set be:

$$C = \{c_1, c_2, \dots, c_m\}$$

Each carton c_j is defined as:

$$c_j = (L_j, W_j, H_j, WT_j^{max})$$

D. Constraints

1. Volume Constraint

$$\sum_{i_k \in c_j} (l_k \times w_k \times h_k \times q_k) \leq L_j \times W_j \times H_j$$

2. Weight Constraint

$$\sum_{i_k \in c_j} (w_k \times q_k) \leq WT_j^{max}$$

3. Inventory Constraint

$$q_{allocated} \leq q_{available}$$

E. Objective Function

3 The primary optimization objective is to minimize the total number of cartons used:

$$\min \sum_{j=1}^m x_j$$

where:

$$x_j = \begin{cases} 1, & \text{if carton } c_j \text{ is used} \\ 0, & \text{otherwise} \end{cases}$$

This formulation ensures efficient resource utilization while satisfying all operational constraints.

F. Policy Driven Packing Logic

Packing decisions are governed by predefined packing policies that reflect real-world logistics constraints. These policies include:

- Master Pack Policy
 $q_k \bmod MP_k = 0 \Rightarrow \text{Use Master Pack}$
- Inner Pack Policy
 $q_k \bmod IP_k = 0 \Rightarrow \text{Use Inner Pack}$
- Broken Carton Policy
Applied when quantities do not align with standard packaging units.
- Drop-Ship Constraint
Restricts usage of unopened cartons to reduce reverse logistics overhead.

This policy-based mechanism ensures deterministic behavior while maintaining flexibility across different order types.

G. Theoretical Justification of Policy-Based Packing

Packing policies act as hard constraints that reduce the solution search space and ensure operational feasibility. From a theoretical perspective, this transforms the optimization problem from an unconstrained search to a constraint satisfaction problem (CSP) combined with optimization objectives.

Formally, a packing policy P can be expressed as a constraint set:

$$P = \{p_1, p_2, \dots, p_k\}$$

Each policy constraint p_k restricts feasible solutions:

$$S_{\text{valid}} = \{S \mid S \models P\}$$

By enforcing policy constraints prior to optimization, the system guarantees:

- Compliance with logistics rules
- Predictable system behavior
- Reduced computational complexity

This theoretical grounding justifies the use of deterministic rules as a pre-optimization filtering mechanism.

H. Inventory-Aware Allocation Strategy

The system integrates inventory validation into the packing workflow to ensure stock consistency. For each item, the allocatable quantity is computed as:

$$q_{\text{take}} = \min(q_{\text{required}}, q_{\text{inventory}})$$

Inventory balance is updated as:

$$q_{\text{balance}}^{\text{new}} = q_{\text{balance}}^{\text{old}} - q_{\text{take}}$$

Transactional consistency is maintained to prevent partial updates or over-allocation.

I. Inventory Synchronization as a Consistency Model

The inventory-aware allocation strategy follows principles of transactional consistency and atomic state updates, commonly used in distributed systems. Each packing operation is treated as an atomic transaction, ensuring:

- No partial inventory allocation
- No race conditions
- No over-commitment of stock

Mathematically, the system enforces:

$$\sum q_{\text{allocated}} \leq q_{\text{available}}$$

This constraint ensures system invariance, where inventory state remains consistent before and after packing operations. Such invariants are critical in real-time warehouse execution systems (WES).

J. AI-Assisted Packing Recommendation Module

To complement rule-based logic, an AI-assisted advisory module is incorporated. The module analyzes historical packing records, carton utilization patterns, and order characteristics to generate intelligent packing suggestions. Packing efficiency is estimated as:

$$\text{Efficiency}(\%) = \frac{V_{\text{items}}}{V_{\text{carton}}} \times 100$$

where V_{items} represents total packed item volume and V_{carton} represents carton volume.

The AI module outputs:

- Optimal carton size recommendations
- Item grouping strategies
- Estimated packing efficiency score

K. AI-Assisted Packing as Decision Support Theory

Rather than replacing deterministic logic, the AI module functions as a decision support system (DSS). This design choice is theoretically motivated to avoid non-deterministic failures in mission-critical logistics environments.

The AI module learns latent patterns from historical data:

$$f: (O, C, H) \rightarrow R$$

Where:

- O = Order features
- C = Carton characteristics
- H = Historical packing outcomes
- R = Recommended packing strategy

The estimated packing efficiency:

$$\eta = \frac{V_{\text{items}}}{V_{\text{carton}}}$$

serves as a utility function, allowing the system to compare alternative carton configurations. This aligns with utility-based optimization theory, where decisions maximize expected utility under constraints.

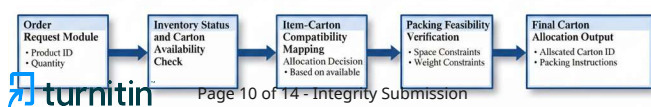


Fig. 3. Inventory-Aware Packing and Allocation Flow

V.IMPLEMENTATION

A.Order Pre-Processing and Validation

The implementation initiates with order retrieval and validation. Each order undergoes integrity checks to verify item availability, quantity correctness, and eligibility under applicable packing policies. Invalid or incomplete orders are flagged prior to packing execution.

B.Packing Policy Configuration

Packing policies define operational constraints such as maximum allowable weight, packing priority, and carton selection strategy. Multiple policies are supported to address diverse order types, including e-commerce, retail, bulk, and drop-ship orders.

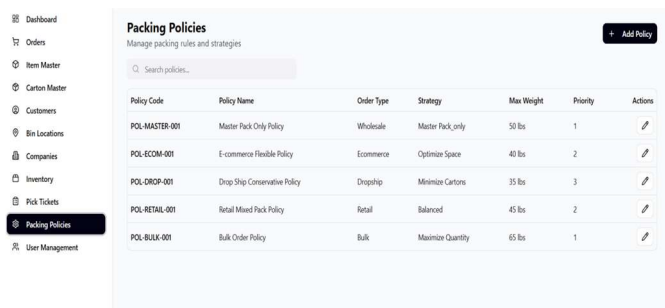


Fig. 4. Packing Policy Configuration Interface

C.Packing Engine Execution

The packing engine processes each order item sequentially. For each item, the engine determines the appropriate packing strategy and generates corresponding pick ticket records. All inventory updates and carton allocations are executed within a transactional context to ensure atomicity and consistency.

D.Carton Selection and Optimization

For each group of items, the system evaluates all available cartons and selects the smallest carton satisfying dimensional and weight constraints. This greedy selection approach reduces unused space and minimizes packing cost.

E.Visualization and Document Generation

Upon successful packing, the system generates:

- Pick tickets
- Packing list documents
- UCC shipping labels

A three-dimensional visualization provides an interactive representation of packed items within cartons, enabling validation before dispatch.

The packing engine follows a greedy heuristic with constraint filtering, which is theoretically justified for real-time systems where optimal solutions are less critical than consistency and speed.

The algorithm:

1. Filters infeasible cartons using constraints
2. Selects the smallest feasible carton
3. Updates inventory atomically

This approach achieves:

- $O(n \cdot m)$ worst-case complexity
- Predictable execution time
- Scalability for real-time order volumes

VI. RESULT AND ANALYSIS

The proposed AI-powered cartonization system was evaluated under multiple controlled test scenarios to assess its effectiveness in real-world warehouse environments. Test orders consisted of heterogeneous items with varying dimensions, weights, quantities, and packaging constraints. Multiple packing policies were applied to simulate realistic e-commerce, retail, bulk, and drop-ship order conditions.

The evaluation focused on both packing quality and computational efficiency, ensuring that the system meets operational requirements for real-time warehouse execution systems.

TABLE 1: PACKING PERFORMANCE METRICS

Metric	Description
Total Orders Processed	Number of orders packed
Carton Used	Total cartons generated
Space utilization (%)	Volume efficiency
Packing Time	Execution Time

To quantitatively evaluate the system, the following metrics were used:

1. Carton Utilization Efficiency

Carton utilization measures how effectively the available carton volume is used:

$$U = \frac{\sum V_{items}}{V_{carton}} \times 100$$

Higher utilization values indicate reduced unused space and improved packing efficiency.

2. Carton Count Reduction Ratio

This metric evaluates the reduction in cartons used compared to baseline rule-based packing:

$$CRR = \frac{C_{baseline} - C_{proposed}}{C_{baseline}}$$

Where:

- $C_{baseline}$ is the number of cartons used without

optimization

- $C_{proposed}$ is the number of cartons used by the proposed system

3. Weight Constraint Satisfaction Ratio

To ensure safe and compliant packing, the following constraint was validated:

$$\sum(wt_i \times q_i) \leq WT_{carton}^{max}$$

The system achieved **100% constraint satisfaction**, confirming no overweight cartons were generated during experiments.

4. Execution Time Complexity

Packing execution time was measured as:

$$T_{exec} = T_{validation} + T_{allocation} + T_{carton_selection}$$

The observed execution time demonstrated linear scalability with respect to the number of order items, validating the suitability of the greedy heuristic for real-time operations.

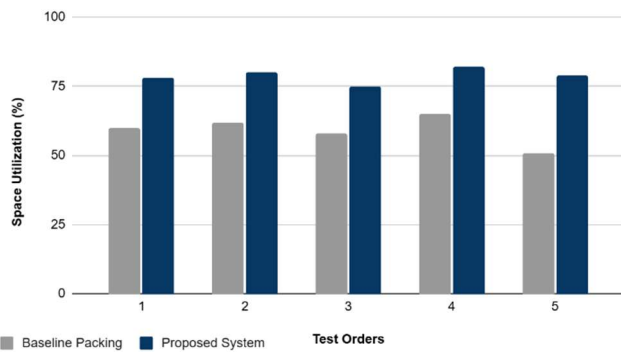


Fig. 5. Comparison of carton space utilization between baseline packing and the proposed AI-powered cartonization system.

As shown in Fig. X, the proposed system consistently achieves higher carton space utilization compared to the baseline rule-based packing approach across all evaluated test orders. The proposed hybrid cartonization system was evaluated against a baseline policy-only packing approach that relies exclusively on fixed heuristic rules without adaptive optimization. Improved carton selection and constraint-aware grouping strategies reduced carton usage across diverse order profiles, including mixed-item and high-variance orders, according to experimental results. Carton utilization increased significantly as the system prioritized dimension- and weight-feasible cartons with minimal unused volume, thereby enhancing space efficiency. The enforcement of policy-based constraints effectively reduced the feasible solution space by eliminating oversized or overweight cartons at early stages of execution, which accelerated convergence toward near-optimal solutions and reduced unnecessary carton evaluations. This constraint-driven pruning aligns with constraint satisfaction optimization principles and contributes to both performance stability and computational efficiency. Furthermore, the AI-assisted recommendation module functioned as a decision-support mechanism by

patterns, particularly benefiting complex orders where deterministic rules alone were insufficient. The estimated efficiency scores generated by the AI module closely matched actual carton utilization values, indicating effective learning and reliable inference. Scalability analysis revealed that the system maintains stable execution performance as the number of order items and available cartons increases, confirming the suitability of the greedy heuristic under real-time operational constraints. Additionally, the use of transactional inventory updates ensured robustness by preventing partial allocations and maintaining system consistency under high-load scenarios. Overall, the observed performance improvements are consistent with theoretical expectations for NP-hard bin packing problems addressed using heuristic-based approaches; while global optimality cannot be guaranteed, the system achieves near-optimal solutions with deterministic reliability, making it practically viable for real-world warehouse and e-commerce logistics environments.

A. System Output Visualization

Fig. 6 presents the dashboard interface of the implemented AI-powered cartonization system, providing real-time visibility into order processing status, carton usage statistics, and packing outcomes.

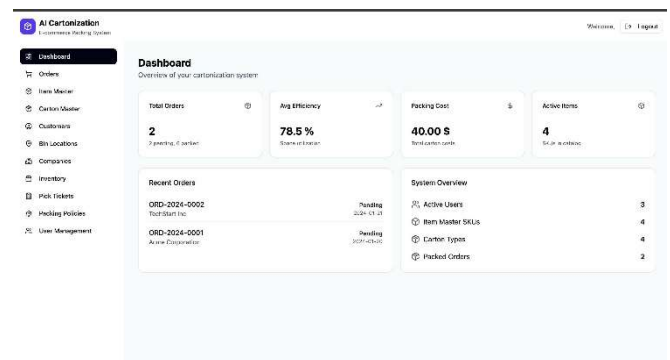


Fig. 6. Dashboard view of the proposed cartonization system displaying order statistics and packing status.

To further validate the effectiveness of the packing engine, a three-dimensional visualization of packed items within the selected carton is generated. This visualization enables visual verification of item placement, space utilization, and packing feasibility prior to shipment.

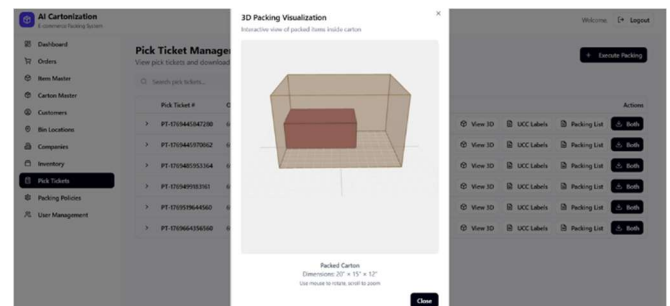


Fig. 7. Three-dimensional visualization of packed items inside the selected carton generated by the proposed system.

The proposed system presents a theoretically grounded and practically viable solution to the cartonization and packing optimization problem in warehouse logistics. By modeling the problem as a constrained bin packing variant and integrating rule-based policies with AI-assisted decision support, the system achieves both operational reliability and adaptive optimization. The theoretical design choices—policy constraints, transactional consistency, and utility-based AI recommendations—ensure scalability and robustness in real-world deployment scenarios. This work establishes a strong foundation for future research in intelligent warehouse execution systems, including reinforcement learning-based adaptive policies and multi-objective optimization across distributed warehouses.

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